Chapter 9

Medical Mycology

Learning Objectives

After studying this chapter the students will be able to,

- *Identify the pathogenic fungi most commonly causing disease by using advanced techniques. Fungal infection is common in developing countries so we should be aware to prevent and treat the fungal infection.*
- *Study the taxonomy, structure and classification of medically important fungi.*
- *Study about the mycoses and its pathogenesis, clinical feature, and its treatment and prophylaxis.*
- *Study about the collection, processing of the sample and its molecular diagnosis.*

Chapter Outline

9.1 Classification of fungi based on the Host parasitic Relationship

9.2 Superficial Cutaneous Mycosis

9.3 Subcutaneous Mycoses

9.4 Systemic Mycosis

9.5 Opportunistic Mycosis

5N9S8R

The branch of biology that deals with the study of fungi is known as **"Mycology"**. The term is derived from Greek work **'Mykes'** means **mushroom** and **'Logos'** means **study**. Medical Mycology is the study of fungal epidemiology, ecology, pathogenesis, diagnosis and treatment in human beings**. Raymond Jacques Sabouraud (1864–1936)** is the father of Medical Mycology.

9.1 Classification of Fungi based on the Host Parasitic Relationship

Based on the host parasitic relationship the fungi are grouped into three types.

a. Commensalism: The fungus neither gets benefit nor harmed by the host parasitic relationship.

b. Mutualism: The fungus benefited from the host parasitic relationship.

c. Parasitism: The host is harmed by the fungus in host parasitic relationship.

Based on their wide spectrum of adaptability, fungi causing human mycoses can be categorized into:

a. Pathogenic fungi: The ability of the fungi to adapt to skin flora and cause infection.

TN_GOVT_XII_Micro_Biology_CH09.indd 188 27-02-2019 14:55:12